Repair Histories: Model 3 Battery Work Orders (SYNTHETIC TEST DOCUMENT)

> **SYNTHETIC TEST DATA.** All work orders, vehicles, workshops, dates and outcomes are invented to test the FixSkrew pipeline. They do not describe any real vehicle or repair.

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Each entry gives the complaint, the diagnosis, the repair, the verification and the outcome. Outcomes are recorded so the assistant can tell which repairs resolved a fault and which did not.

RO-1001: Weak cell group, module replaced

Vehicle: Model 3 Standard Range Plus, 2020, 96,400 km · **Opened:** 2025-02-05 **Complaint:** Range loss and charging stopping early. **Codes:** P0B3C active, P0A7F stored.

Diagnosis. Cell spread 0.140 V at rest. A balancing charge did not recover it. Group 63 of module 3 dropped 0.188 V more than its neighbours under 150 A. The module 3 sense connector was reseated and inspected: no corrosion, readings unchanged.

Repair. Module 3 replaced. BMS capacity learning routine run (full discharge, full charge at a low rate).

Verification. Cell spread at rest 0.012 V. Under 150 A load the spread was 0.071 V. Full charge reached 100 % without early stop. P0B3C did not return after 3 wake cycles and a 200 km drive.

Outcome. Resolved. Range recovered by about 14 % compared with before the repair.

Lesson. The module was replaced only after the weak group was confirmed by measurement under load and after the connector was excluded.

RO-1002: False cell imbalance from a corroded sense connector

Vehicle: Model 3 Long Range, 2019, 142,500 km · **Opened:** 2025-02-18 **Complaint:** Reduced range and a warning about the battery. **Codes:** P0B3C active.

Diagnosis. Group 29 of module 2 read 3.120 V while its neighbours were at 3.830 V, a spread of 0.710 V. A reading this low in a pack with normal behaviour was suspect. Inspection found green corrosion on the sense connector pin of module 2.

Repair. Connector cleaned, pin re-crimped and sealed with dielectric grease. No module replaced.

Verification. Group 29 now read 3.826 V. Spread 0.016 V. The code did not return.

Outcome. Resolved without parts. A module replacement would have been wrong.

Lesson. A very large single-group deviation can be a sense wiring fault, not a cell fault.

RO-1003: Imbalance cured by balancing charge

Vehicle: Model 3 Standard Range Plus, 2021, 30,200 km · **Opened:** 2025-03-04 **Complaint:** "Range is lower than before, and charging slows near full." **Codes:** P0B3C stored.

Diagnosis. Spread 0.062 V at rest. The owner always kept the car between 30 % and 70 %, as described in SYN-TSB-24-010. Pack temperature and insulation were normal.

Repair. Balancing charge: 100 % on AC, 4 hours held, 2 hours rest.

Verification. Spread dropped to 0.011 V. The code was cleared and did not return.

Outcome. Resolved. Advice given to charge to 100 % once a month.

RO-1004: Coolant leak into the pack, cooling plate replaced

Vehicle: Model 3 Long Range, 2021, 61,000 km · **Opened:** 2025-03-25 **Complaint:** HV warning after heavy rain. **Codes:** P0AA6 stored (3 times).

Diagnosis. Pack insulation 410 Ω/V with all loads disconnected. Coolant level low with no external leak. Pressure test showed a leak at a cooling plate port. Campaign SYN-RC-24-002 had not been performed on this vehicle.

Repair. Pack removed. Enclosure dried, port seal and cooling plate replaced, pack vent replaced, coolant refilled and bled.

Verification. Insulation 1,780 Ω/V after reinstall and 1,750 Ω/V after a drive and a fast charge. Coolant level stable over two weeks. No codes.

Outcome. Resolved. Campaign recorded as completed.

RO-1005: Isolation fault from a worn charge-port harness, not the pack

Vehicle: Model 3 Standard Range Plus, 2020, 88,000 km · **Opened:** 2025-04-02 **Complaint:** Isolation warning that appears after washing the car. **Codes:** P0AA6 stored.

Diagnosis. With the pack alone, insulation was healthy (1,500 Ω/V). Reconnecting branches one by one showed a drop to 280 Ω/V when the charging inlet branch was connected. Water traces were found at the inlet HV connector.

Repair. Charging inlet HV harness section and seal replaced.

Verification. Insulation 1,480 Ω/V with every branch connected. Car washed twice, no recurrence.

Outcome. Resolved. No pack work needed.

Lesson. An isolation fault does not always sit in the pack. Test branch by branch.

RO-1006: Interlock fault, connector latch replaced

Vehicle: Model 3 Long Range, 2019, 118,000 km · **Opened:** 2025-04-09 **Complaint:** Lost drive power on a rough road. **Codes:** P0A0D stored (4 occurrences).

Diagnosis. Pack healthy. Junction box connector latch loose, campaign sticker absent. Matches SYN-RC-24-001.

Repair. Campaign SYN-RC-24-001 carried out: retention clip replaced, pins inspected, locking tab fitted.

Verification. Test drive over rough road, no recurrence in 300 km. Code cleared.

Outcome. Resolved.

RO-1007: End-of-life pack, modules replaced

Vehicle: Model 3 Standard Range Plus, 2019, 187,000 km · **Opened:** 2025-05-08 **Complaint:** Shrinking range, only 220 km. **Codes:** P0A7F active.

Diagnosis. Uniform ageing: state of health 68 % after software update and relearn, measured capacity 36.8 kWh, small cell spread. Not a single weak group.

Repair. Owner chose full pack replacement with a refurbished unit. New pack coded to the vehicle.

Verification. State of health 97 %. Capacity test passed. Fast charge and range test normal.

Outcome. Resolved. Old pack returned to the recycler.

Lesson. Replacing one module would not have restored the range, because all modules were aged.

RO-1008: Pre-charge resistor open, junction box replaced

Vehicle: Model 3 Long Range, 2019, 134,000 km · **Opened:** 2025-06-18 **Complaint:** Slow start and loud clunk. **Codes:** P0AE2 active.

Diagnosis. 12 V battery healthy at 12.6 V. Pre-charge resistor open circuit, pre-charge time 3.8 s.

Repair. HV junction box assembly replaced.

Verification. Pre-charge time 1.1 s. No codes in 20 wake cycles.

Outcome. Resolved.

RO-1009: False contactor fault from a weak 12 V battery

Vehicle: Model 3 Standard Range Plus, 2020, 70,500 km · **Opened:** 2025-06-30 **Complaint:** Intermittent HV warning and contactor codes after the car sat for a week. **Codes:** P0ADD stored, P0AE2 stored.

Diagnosis. 12 V battery at 10.4 V after sitting. The contactors and pre-charge resistor measured within specification.

Repair. 12 V battery replaced. No HV parts replaced.

Verification. Codes cleared. No recurrence after two weeks of mixed use.

Outcome. Resolved.

Lesson. Check the 12 V supply first when contactor codes are intermittent. Replacing the junction box would have been unnecessary.

RO-1010: Charge port sensor wiring, harness replaced

Vehicle: Model 3 Long Range, 2020, 72,000 km · **Opened:** 2025-07-03 **Complaint:** Charging stops after minutes with a charge port message. **Codes:** P1A00 stored.

Diagnosis. Reported port temperature 88 °C while the surface was 26 °C. Sensor circuit open intermittently near the connector. Matches SYN-RC-23-007.

Repair. Sensor harness and connector replaced. The charge port itself was kept.

Verification. Reported temperature tracked the surface temperature during a 2-hour AC charge and a DC session.

Outcome. Resolved.

RO-1011: Pump failure causing thermal limits

Vehicle: Model 3 Standard Range Plus, 2021, 49,000 km · **Opened:** 2025-08-20 **Complaint:** Reduced power, warning, slow charging. **Codes:** P0A81 active, P0A9C active, P0B3C stored.

Diagnosis. Pump commanded to 100 % but not turning. Pump fuse intact. Pump connector corroded. Module 4 temperature of 58 °C was checked with a contact thermometer: 33 °C, so the sensor reading was false. Cell spread was re-measured after the cooling repair.

Repair. Coolant pump and connector replaced. Module 4 temperature sensor connector reseated. Coolant bled.

Verification. Pump speed matched the command. All module temperatures agreed within 1.5 °C. Cell spread 0.013 V after the cooling repair. No codes.

Outcome. Resolved. The stored P0B3C was a consequence of the thermal fault, not a bad cell.

Lesson. Fix the thermal fault first, then re-measure cell spread before judging a module.

RO-1012: Unnecessary module replacement (negative outcome)

Vehicle: Model 3 Standard Range Plus, 2020, 54,000 km · **Opened:** 2025-09-02 **Complaint:** Range loss. **Codes:** P0A7F stored.

Diagnosis (original). A single reading showed state of health 79 %. A module was replaced on the strength of that value alone.

Outcome. Not resolved. State of health stayed at 79 % after the replacement. A later check showed that the BMS firmware was out of date and the estimate was incorrect. After the software campaign SYN-TSB-24-018 and a relearn cycle the state of health was 93 %.

Lesson. A single state-of-health reading must not be used to condemn a module. Update software and verify capacity by a test first.

RO-1013: Cold-weather charge complaint closed as normal

Vehicle: Model 3 Standard Range Plus, 2022, 24,900 km · **Opened:** 2025-01-30 **Complaint:** Slow DC charging in winter. **Codes:** none.

Diagnosis. Charge power rose from 22 kW to over 100 kW as the pack warmed. Heater and pump normal.

Repair. None. Owner advised to precondition before arrival at the charger (SYN-TSB-24-014).

Outcome. Closed with no fault found. Owner confirmed shorter sessions after preconditioning.

RO-1014: Recurrence after an incomplete repair

Vehicle: Model 3 Long Range, 2021, 66,500 km · **Opened:** 2025-10-01 **Complaint:** Isolation warning returned 6 weeks after a previous repair. **Codes:** P0AA6 stored.

Diagnosis. The earlier repair replaced the coolant pump and refilled the coolant, but did not pressure-test the cooling plate. Pressure test now showed a leak at the cooling plate port. Campaign SYN-RC-24-002 was still open.

Repair. Cooling plate and port seal replaced and pack dried, vent replaced.

Verification. Insulation 1,700 Ω/V, stable after fast charging and rain exposure.

Outcome. Resolved at the second attempt.

Lesson. When coolant is lost together with an isolation fault, pressure-test the plate before replacing smaller parts.

Summary of outcomes

Order	Fault	Repair	Outcome
RO-1001	Weak group, module 3	Module replaced	Resolved
RO-1002	Corroded sense connector	Connector repaired	Resolved
RO-1003	Imbalance from partial charging	Balancing charge	Resolved
RO-1004	Coolant leak into pack	Cooling plate replaced	Resolved
RO-1005	Water at the charge inlet harness	Inlet harness replaced	Resolved
RO-1006	Interlock connector latch	Campaign applied	Resolved
RO-1007	End-of-life pack	Pack replaced	Resolved
RO-1008	Open pre-charge resistor	Junction box replaced	Resolved
RO-1009	Weak 12 V battery	12 V battery replaced	Resolved
RO-1010	Charge port sensor wiring	Harness replaced	Resolved
RO-1011	Pump failure	Pump replaced	Resolved
RO-1012	Module replaced on one reading	none useful	Not resolved
RO-1013	Cold-weather charging	none	No fault
RO-1014	Incomplete earlier repair	Cooling plate replaced	Resolved